



Insect Bites

Factors to Consider

The bites of insects and other arthropods may be a minor nuisance or may lead to serious medical problems, including transmission of insect-borne illnesses and severe allergic reactions.



(VOCs) emanating from human skin.

Mechanisms of Injury: Arthropod bites consist of punctures made by the mouth.

Mouthpart Types: There are two methods of obtaining blood by arthropods. Some groups, such as mosquitoes, bed bugs, kissing bugs, and sucking lice, obtain blood directly from capillaries or small veins of the offending organism.

Others, such as ticks, horse flies and deer flies, black flies, and biting midges, obtain their blood by lacerating blood vessels and feeding from the resulting pool of blood.

Host (Human) Factors: There is some evidence that specific host factors may predispose individuals to increased arthropod biting. Analyses of the distributions of transmission rates of insect-borne infections indicate that, typically, 20 percent of the host (human) population is [responsible for approximately 80 percent of disease transmission](#) by blood-sucking insects.

Possible factors that attract mosquitoes specifically include blood type, metabolic rate, amount of CO₂ released, body temperature, clothing types and colors, and varying amounts of volatile organic compounds

Arthropods That Bite Humans

Arthropods That Commonly Bite Humans				
Mosquitoes	Ticks	Kissing bugs	Bed bugs	Black flies
Horse and deer flies	Sand flies	Stable flies	Biting midges	Fleas
Centipedes	Biting mites	Chiggers	Lice	A small number of types of spiders

Types of Reactions and Treatments

Insect bites may result in several types of local reactions, papular urticaria, or systemic allergic reactions. Rarely, other forms of systemic reactions occur, [such as serum sickness](#).

Local Reactions

The normal reaction to an insect bite is an inflammatory reaction at the site of the punctured skin, which appears within minutes and consists of pruritic local erythema and edema. A single punctum may be visible. Dual puncta can be seen with spider bites and centipede envenomations. Symptoms usually subside within a few hours.

Local reactions are [caused by irritant substances concentrated in insect saliva](#) (e.g., anticoagulants such as factor Xa inhibitors, digestive enzymes such as amylases and esterases, agglutinins, and mucopolysaccharides). In some cases, a local reaction is followed by a delayed skin reaction, consisting of local swelling, itching, and redness.

Treating Local Reactions

Insect bites and local reactions should be washed with soap and water. Reduction of local edema may be induced with cooling (ice or cold pack). Topical creams, gels, and lotions, such as those containing calamine or [pramoxine](#), can be helpful in reducing pruritus, if necessary. However, routine use of topical anesthetic and antihistamine preparations [should be avoided](#) because they can sensitize the skin following sun exposure and induce allergic contact sensitivity.

Minimally sedating oral antihistamines (e.g., [cetirizine](#) or [fexofenadine](#), once or twice daily) may be helpful for patients with troublesome itching and are preferred over sedating agents, particularly in small children. However, the sedating agent hydroxyzine (hydrochloride or pamoate; 10 to 25 mg every four to six hours, as needed) may be helpful for controlling persistent pruritus in adults, perhaps largely due to the sedating effects. H1 and H2 antihistamines (e.g., [famotidine](#)) may be used concurrently. Oral antihistamines should not be used for children under 6 months of age.

Dramatic local swelling and induration can be reduced with a brief course of oral glucocorticoids, although this should be reserved for severe cases.

Unusual Local Reactions

- Uncommonly, local reactions evolve to become vesicular ([picture 1](#)), bullous ([picture 2](#) and [picture 3](#)), or necrotic ([picture 4](#)). Patients should maintain good hygiene and avoid scratching to prevent infection.
- Bullous reactions have been reported with bed bugs and other arthropods. Bullae should be kept clean as they heal, and, if lesions persist, [they may be treated with topical corticosteroids](#). They may warrant additional workup to rule out other bullous disorders. If the arthropod bite has resulted in longstanding bullous lesions with tissue eosinophilia, [the diagnosis of eosinophilic cellulitis should be considered](#).
- Necrotic lesions most commonly result from the injection of venom, which occurs with [some spider bites](#) and centipede bites.

Papular Urticaria

Papular urticaria is a hypersensitivity disorder in which insect bites, most often those of fleas, mosquitoes, or bed bugs, lead to recurrent and sometimes chronic, itchy papules on exposed areas of skin (e.g., arms, lower legs, upper back, scalp). The 0.5 to 1 cm lesions may be urticarial at the start of the syndrome but become persistent and papular and/or nodular with time.



Note. From *Papular Urticaria* [Photograph], by Aye Min Htoo, n.d., Dr. Aye Min Htoo (<https://drayeminhtoo.com/papular-urticaria/>).

Treating Papular Urticaria

Management of papular urticaria includes selective and limited use of nonsedating antihistamines for pruritus, mid-potency topical corticosteroids applied to individual lesions, and reassurance, as the [disorder eventually resolves spontaneously](#).

Systemic Allergic Reactions

Systemic allergic reactions to insect bites are uncommon.

Treating Systemic Allergic Reactions

Anaphylaxis associated with insect bites needs to be treated promptly with [epinephrine](#). Patients who have experienced systemic reactions should be supplied with an epinephrine autoinjector or nasal spray and instructed in how and when to use it.

Preventing Systemic Allergic Reactions

Guidelines on vector control on a global level, focus on reducing or eliminating larval habitats. On an individual level, [patients should be instructed to apply insect repellents](#) that contain active ingredients such as N,N-diethyl-3-methylbenzamide (DEET), oil of lemon eucalyptus, para-menthane-diol (PMD), or 2-undecanone. Additional protective measures include the use of protective covers, such as screens on windows and netting over beds.

Flea Bites

Fleas inflict bites that are often inconsequential and ignored. They can, however, result in local reactions or even eschars, pustules, or necrotic lesions.

Flea Bite Reactions

Flea bites are usually papules arranged in a nonfollicular pattern, but can induce papular urticaria.

Flea Bite Prevention

Fleas can be deterred on humans by DEET and other insect repellents. On animals, there are a variety of effective veterinary products that may be directly applied to pets and domestic animals.

Mite Bites

Mites are tiny arachnids that display incredible diversity in form and habitat. Many species of mites may bite people, including chigger mites; rat, chicken, and fowl mites; straw itch mites; scabies mites; cheyletid mites; bird mites, and others. Apart from scabies mites and follicle mites, all cases of biting mites are self-limiting and will subside after removing the source of infestation (i.e., they do not take up residence and live on humans).

Mite Bite Reactions

Bites from mites are generally small, erythematous papular lesions that are intensely pruritic. They may have a wheal appearance or appear pustular, vesicular, or, rarely, bullous. They can become crusted and need to be monitored for progression to secondary infection. Both chigger bites and scabies are generally localized or grouped, though scabies have burrows as a typical finding.

Mite Bite Prevention

People complaining about mites biting them should be evaluated for exposure to mite-infested areas and a detailed history obtained to rule out exposure to infested animals or (certain) grains and food products. In addition, persons claiming to be infested with mites [may be suffering from delusions of parasitosis](#). Chiggers can be deterred by DEET or by wearing knee-high rubber boots when outdoors in infested areas.



Note. From *Flea Bites* [Photograph], by Unbound Medicine, n.d., Unbound Medicine (https://www.unboundmedicine.com/5minute/view/5-Minute-Clinical-Consult/1688537/all/Arthropod_Bites_and_Stings).

References

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